

UNIVERSITY COLLEGE LONDON

EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : **COMP0078**

ASSESSMENT PATTERN: **COMP0078A7PF 001**

MODULE NAME : **COMP0078 - Supervised Learning**

LEVEL : **Postgraduate**

DATE: : **02/05/2024**

TIME : **14:30**

DURATION : **3 Hours**

This paper is suitable for candidates who attended classes for this module in the following academic year(s):

**Year
2023/24**

**EXAMINATION PAPER CANNOT BE REMOVED FROM THE EXAM HALL. PLACE
EXAM PAPER AND ALL COMPLETED SCRIPTS INSIDE THE EXAMINATION
ENVELOPE**

Hall instructions	N/A
Additional materials	N/A
Standard Calculators	Yes
Non-Standard Calculators	No

TURN OVER

Suitable for Cohorts: 23/24

Answer ALL FIVE (5) questions.

Marks for each part of each question are indicated in square brackets.

Standard calculators are permitted.

1. **(Variants of Least Squares)** A common preprocessing in machine learning is to center the data. In this problem we will see how this can be related to working with an (unpenalized) offset term in the case of linear functions.

- a. Consider Tikhonov regularization in the linear case, but assume that there is an unpenalized offset term b ,

$$\min_{w \in \mathbb{R}^d, b \in \mathbb{R}} \left\{ \frac{1}{n} \sum_{i=1}^n (\langle w, x_i \rangle + b - y_i)^2 + \lambda \|w\|^2 \right\}, \quad (1)$$

and let (w^*, b^*) be the solution of the above problem. Denote by $x_i^c = x_i - \bar{x}$, $y_i^c = y_i - \bar{y}$ the centered data for $i = 1, \dots, n$, where $\bar{y}, \bar{x}$ are the output and input means respectively.

- i. Write the problems in matrix notation as done in the course slides (e.g. with input points collected in a single matrix $X \in \mathbb{R}^{n \times d}$, the outputs in a vector $y \in \mathbb{R}^n$ etc.)

[1 marks]

- ii. Show that w^* also solves

$$\min_{w \in \mathbb{R}^d} \left\{ \frac{1}{n} \sum_{i=1}^n (\langle w, x_i^c \rangle - y_i^c)^2 + \lambda \|w\|^2 \right\}. \quad (2)$$

[5 marks]

[Question 1 cont. over page]

[Question 1 cont.]

iii. Give closed form expressions for the solutions b^* and w^* .

[3 marks]

iv. Can you “kernelize” the method? Namely, can you find parameters $c_1, \dots, c_n \in \mathbb{R}$ such that

$$\langle w^*, x \rangle = \sum_{i=1}^n \langle x_i, x \rangle c_i$$

for any $x \in \mathbb{R}^d$? Note, at this point you can “substitute” the inner product $\langle x_i, x \rangle$ with $k(x_i, x)$ where $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ a reproducing kernel.

[3 marks]

b. In binary classification settings with imbalanced data (i.e., when there are many more examples of one class than of the other) a common strategy is to *weight* the loss function so that the errors in one class are counted more than those of the other class. In the case of linear regularized least squares, this corresponds to solving the modified problem

$$\min_{w \in \mathbb{R}^d} \left\{ \sum_{i=1}^n \gamma_i (\langle w, x_i \rangle - y_i)^2 + \lambda \|w\|^2 \right\},$$

where the weights $\{\gamma_i : i = 1, \dots, n\}$ are chosen so that $\sum_{i=1}^n \gamma_i = 1$ and $\gamma_i > 0$ for all $i = 1, \dots, n$.

i. Write the problem above in matrix notation.

[1 marks]

ii. Derive the explicit form of the minimizer w^* .

[2 marks]

iii. Consider the case of a weighted loss function *and* an unpenalized offset $b \in \mathbb{R}$,

$$\min_{w \in \mathbb{R}^d, b \in \mathbb{R}} \left\{ \sum_{i=1}^n \gamma_i (\langle w, x_i \rangle + b - y_i)^2 + \lambda \|w\|^2 \right\}$$

Derive the explicit form of the minimizers w^*, b^* for this problem, using the results derived above.

[5 marks]

2. Adaboost

- a. Derive Adaboost as the greedy minimizer of empirical risk

$$\min \left\{ \frac{1}{n} \sum_{i=1}^n \ell(f_T(x_i), y_i) \mid f_T = \sum_{t=1}^T \alpha_t h_t, \alpha_1, \dots, \alpha_T \in \mathbb{R}, h_1, \dots, h_T \in \mathcal{H} \right\},$$

with $\mathcal{H}$ a space of hypotheses.

In particular:

- i. Which loss function ℓ yields the Adaboost algorithm?
- ii. What kind of greedy strategy is used.
- iii. Explicitly write down the update rule to transition from f_t to f_{t+1}

[10 marks]

- b. In class we have discussed that it is usually a problem to minimize the empirical risk over a class of hypotheses that is too large or expressive (overfitting). Is this also the case for Adaboost? If yes, are there straightforward strategies to add regularization to Adaboost?

[5 marks]

3. First Order Optimization + Kernels

- a. Recall the definition of M -smoothness for a function $F : \mathbb{R}^d \rightarrow \mathbb{R}$.

[1 marks]

- b. Assume to have a loss ℓ that is differentiable and M -smooth (uniformly in the first variable) and a training set $(x_i, y_i)_{i=1}^n$ with $x_1, \dots, x_n \in \mathbb{R}^d$. Recall how Gradient Descent (GD) is defined to minimize the empirical risk $\mathcal{E}_n(w) = \frac{1}{n} \sum_{i=1}^n \ell(w^\top x_i, y_i)$.

[3 marks]

- c. Show that GD can be “kernelized”: namely, for any iterate w_t of GD, you can find parameters $c_1^{(t)}, \dots, c_n^{(t)} \in \mathbb{R}$ such that

$$\langle w_t, x \rangle = \sum_{i=1}^n \langle x_i, x \rangle c_i^{(t)}$$

for any $x \in \mathbb{R}^d$.

[8 marks]

- d. Discuss how this observation can allow you to apply GD to settings in which you want to use a kernel (possibly associated with an infinite dimensional feature map)

[4 marks]

- e. Comment on how applying GD in the general kernel setting affects the convergence result of GD we discussed in class, possibly providing a result characterizing the convergence rates for GD applied to this setting.

[4 marks]

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4. Rademacher complexity for hard-SVM.

- a. Let $(x_i, y_i)_{i=1}^n$ be a datasets of points $x_i \in \mathbb{R}^d$ and $y_i \in \{-1, 1\}$. Recall the formulation of the hard-SVM problem, (i.e. without slack variables, assuming linear separability of the data points).

[2 marks]

Let $\mathcal{X} = \{x \mid \|x\| \leq R\} \subset \mathbb{R}^d$ for a constant $R > 0$ and $\mathcal{Y} = \{-1, 1\}$. Let ρ be a distribution over $\mathcal{X} \times \mathcal{Y}$ such that there exists a vector $w^* \in \mathbb{R}^d$ such that $\mathbb{P}(yx^\top w^* \geq 1) = 1$. Let $S = (x_i, y_i)_{i=1}^n$ be a datasets of points sampled independently according to $S \sim \rho^n$. Denote $w_S \in \mathbb{R}^d$ the solution to the hard-SVM problem you introduced above.

Use the results derived in class based on the Rademacher complexity, to give an upper bound of the form

$$\mathbb{P}_{(x,y) \sim \rho}(y \neq \text{sign}(x^\top w_S)) \leq O(n^{-1/2})$$

(either in expectation or probability).

Organize your proof along the following points:

- b. Recall the definition of $\ell(x^\top w, y)$ the hinge loss and show that

$$\mathbb{P}_{(x,y) \sim \rho}(y \neq \text{sign}(x^\top w)) \leq \mathcal{E}(w) \quad \text{with} \quad \mathcal{E}(w) = \int \ell(x^\top w, y) d\rho(x, y).$$

(hint: note that the probability of observing discordant signs can be interpreted as the expected risk with respect to the 0 – 1 loss).

[3 marks]

- c. Apply to w_S and $\mathcal{E}$ the result discussed in this course, that allows one to control the excess risk of a learning algorithm taking values in a space $\mathcal{H}$ in terms of the Rademacher complexity of $\mathcal{H}$ (either in expectation or probability).

[7 marks]

[Question 4 cont. over page]

[Question 4 cont.]

- d. Choose the space $\mathcal{H}$ to be the space of linear functions

$$\mathcal{H} = \{f : \mathcal{X} \rightarrow \mathbb{R} \mid f(x) = x^\top w, w \in \mathbb{R}^d \text{ and } \|w\| \leq B\}.$$

Show that there exists a constant $B > 0$ such that for any set S sampled according from ρ^n , the vector w_S has norm $\|w_S\| \leq B$.

[7 marks]

- e. Provide an upper bound for $\mathcal{R}_n(\mathcal{H})$ the Rademacher complexity of $\mathcal{H}$.

[5 marks]

- f. Combine the results above to give a bound on $P_{(x,y) \sim \rho}(y \neq \text{sign}(x^\top w_S))$ either in expectation or probability with respect to $S \sim \rho^n$.

[6 marks]

5. **Spectral Clustering.** Let $L \in \mathbb{R}^{n \times n}$ denote the graph Laplacian of an n -vertex undirected graph.

- a. Show that L has always eigenvector μ_1 with eigenvalue equal to 0. Explicitly provide the form of μ_1 .

[5 marks]

- b. Assume that L admits another eigenvector $\mu_2 \neq \mu_1$, whose corresponding eigenvalue is equal to zero. All other eigenvalues of L are greater or equal than a constant $c > 0$. What can be inferred about the structure of the graph associated to L ? Give the mathematical reasoning behind your inference.

[5 marks]

- c. Explain the role of the second eigenvector when performing spectral clustering with the goal of identifying two clusters

[5 marks]